

Kairn Brannon

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SKILLS:

Python - PyTorch - TensorFlow - SQL - PostgreSQL - Spark/PySpark - Airflow - dbt - MLflow - NLP - AWS - GCP - Azure - Docker - Linux - Splunk - Power BI - Streamlit - Plotly - Django - React - OpenStack

SUMMARY:

ML engineer and data scientist with 7+ years of experience turning messy, real-world data problems into working systems — unifying legacy manufacturing databases, automating reporting pipelines, and building ML models for defense and intelligence clients. Promoted from Principal Data Scientist to AI/ML Manager in under 3 years while staying hands-on, writing code and designing infrastructure alongside my team. Seeking a technical role where I can bring that same builder mentality to hard problems full-time.

EDUCATION:

Stanford Artificial Intelligence Graduate Certificate (anticipated 2026) — 4 graduate-level AI courses, transferable toward Stanford MS

BS Mechanical Engineering — Harvard University, 2017

Deep Learning Specialization (Andrew Ng) [[Certificate](#)] | Applied Data Science Specialization (IBM) [[Certificate](#)]

EXPERIENCE:

Concurrent Technologies Corporation, Washington, DC (Remote)

Manager, AI/ML Jun 2025 – Present

- Manage a team of 8 data scientists delivering AI/ML solutions for DoD and intelligence community clients including the U.S. Army, Marine Corps, NSA, and DARPA
- Contributing to a 3-year adversarial AI research initiative — building attack scripts that generate digital adversarial patches for physical-world testing against autonomous vehicles
- Applying LSTM models to satellite-derived ship movement data to predict future vessel positions for defense applications
- Overseeing AI governance for a DARPA project, establishing data storage and usage standards to inform a framework for future initiatives
- Building ML models on military installation health and safety data — developed 10+ key performance metrics and a Streamlit dashboard for predictions on installation compliance timelines

Senior Principal Data Scientist Jun 2024 – Jun 2025

- Built a centralized relational database unifying 3 welding machines running different software stacks, normalizing data for 2,000+ welds and enabling automated ML pipeline integration
- Spearheaded multiple digital transformation engagements — migrating teams from Excel to full-stack applications (Django, React, PostgreSQL) and Power BI dashboards for 10+ users

Principal Data Scientist Nov 2022 – Jun 2024

- Analyzed cross-domain network traffic for the NSA using Python and Splunk, identifying threat actors attempting to compromise systems between security boundaries
- Built an automated weld analysis framework that processed 2,000+ historical welds and established real-time analysis for future welds, reducing per-weld reporting from ~1 hour to under 5 minutes
- Developed Python-based visualization tools and a Power BI dashboard for energy analysts, automating the generation of congressional reports from Excel data for internal stakeholders

Data Innovation Labs, Washington, DC

Data Scientist Jun 2020 – Nov 2022

- Developed an OCR-based document verification model deployed to beta for client onboarding; managed OpenStack private cloud for 3 client tenants
- Primary client liaison — led discovery calls, demos, and progress updates; explored GPT-3 for automated marketing content generation

Piano Software, New York, NY

Business Development Specialist Oct 2018 – Sep 2019

- Analyzed web traffic data from media publications to build ROI projections for subscription capture; delivered product demos and data analytics presentations to prospects and clients

Earlier Roles

Partnerships Manager — Gooroo (EdTech startup), New York, NY, 2017–2018 | **Lab Engineer** — Harvard Laboratory for Particle Physics and Cosmology, Cambridge, MA, 2014–2017

PROJECTS:

- **Predictive Maintenance ML Pipeline** — End-to-end pipeline predicting turbofan engine RUL (NASA C-MAPSS). Compares RF, XGBoost, and LSTM with interactive Streamlit dashboard.
- **LLM Red Teaming Toolkit** — Adversarial testing toolkit running 32 pre-built attacks across 4 categories, with robustness scoring and risk-rated audit reports. Python/Streamlit/CLI.
- **RAG Research Assistant** — Retrieval-augmented generation app for AI/ML papers using FAISS + hybrid search (semantic + BM25) with page-level citations.
- **Ethereum Validator Infrastructure** — Bare-metal server deployment with 5 Ethereum validators; near-100% uptime for 4 years. Built web app for investment tracking.